

Article

Multimodal Augmented Reality System for Real-Time Roof Type Recognition and Visualization on Mobile Devices

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Abstract: The utilization of augmented reality (AR) is becoming increasingly prevalent in the integration of virtual reality (VR) elements into the tangible reality of the physical world. It facilitates a more straightforward comprehension of the interconnections, interdependencies, and spatial context of data. Furthermore, the presentation of analyses and the combination of spatial data with annotated data are facilitated. This is particularly evident in the context of mobile applications, where the combination of real-world and virtual imagery facilitates enhances visualization. This paper presents a proposal for the development of a multimodal system that is capable of identifying roof types in real time and visualizing them in AR on mobile devices. The current approach to roof identification is based on data made available by public administrations in an open-source format, including orthophotos and building contours. Existing computer processing technologies have been employed to generate objects representing the shapes of building masses, and in particular, the shape of roofs, in three-dimensional (3D) space. The system integrates real-time data obtained from multiple sources and is based on a mobile application that enables the precise positioning and detection of the recipient's viewing direction (pose estimation) in real time. The data were integrated and processed in a Docker container system, which ensured the scalability and security of the solution. The multimodality of the system is designed to enhance the user's perception of the space and facilitate a more nuanced interpretation of its intricacies. In its present iteration, the system facilitates the extraction and classification/generalization of two categories of roof types (gable and other) from aerial imagery through the utilization of deep learning methodologies. The outcomes achieved suggest considerable promise for the advancement and deployment of the system in domains pertaining to architecture, urban planning, and civil engineering.

Keywords: roof types; CNN; deep learning; augmented reality; architecture; urban planning



Academic Editor: Andrea Prati

Received: 22 December 2024

Revised: 17 January 2025

Accepted: 23 January 2025

Published: 27 January 2025

Citation: Kubicki, B.; Janowski, A.; Ingłot, A. Multimodal Augmented Reality System for Real-Time Roof Type Recognition and Visualization on Mobile Devices. *Appl. Sci.* **2025**, *15*, 1330. <https://doi.org/10.3390/app15031330>

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1. Introduction

Traditional maps were designed to simplify the modeling/representation of the real world to facilitate spatial orientation [1,2]. Similarly, contemporary geographical information systems (GIS) serve not only to store spatial data mapping real-world objects employed in a variety of scientific disciplines [3] but also to facilitate the spatial analysis and modeling of natural phenomena [4]. Technological developments allow the use of

virtual reality (VR) or augmented reality (AR) systems for spatial data [5,6]. The use of AR or VR can be found in many fields: in architecture, these tools can be effectively used for interior inventory [7]; in mining, for the visualization of mining infrastructure elements [8], building interior planning [9], and historic preservation [10]; and in medicine, these technologies are used for rehabilitation, mobility assistance for the disabled, and surgical training or treatment of mental illness [11–13]. VR technology allows students to interact with objects outside their normal daily experience, which can increase student engagement and motivation to learn [14]. Historical data have also gained a different way of presenting changes in space, and not only urbanized but also natural changes, thanks to VR [15]. The most common application of VR in the scientific literature is the ability to visualize elements of reality and the fact that the experience is more immersive for the user than traditional methods of presenting data. An interesting engineering solution is the visualization and modeling of urban street flooding based on mobile AR. With such solutions, it is possible to simulate the consequences of disasters and thus increase the awareness of citizens [16].

VR has been identified as a potentially innovative tool to support architects and urban planners in space design and planning. It has been demonstrated that VR can facilitate realistic three-dimensional (3D) visualizations, thereby supporting the analysis of proportions, space layout, and ergonomics of designs. In the context of urban planning, VR has been shown to allow the simulation of the use of public spaces, thus supporting sustainable planning. The technology has been found to increase design precision, shorten decision-making processes, and facilitate communication with stakeholders [17–24].

Despite the numerous applications of VR and AR, there is still a lack of systems that integrate publicly available spatial data in an effective and technically simple way. As a result of the digitization of geodetic data, public administrations (especially those responsible for land use and surveying) have vast amounts of data at their disposal. Much of these data are available in the form of Web Map Service (WMS), Web Feature Service (WFS), and others, or can be downloaded from a dedicated website using the Internet's Hypertext Transfer Protocol (HTTP). Thanks to the services provided by the Head Office of Geodesy and Cartography (GUGiK), data such as orthophotomaps (with a spatial resolution of up to 0.05 m), vector building outlines (with a positional accuracy of 0.10 m), and ALS (Airborne Laser Scanning) LiDAR (Light Detection and Ranging) point clouds, among others, are made available to the public. The photogrammetric products and spatial databases mentioned are used together or individually most often for the purpose of determining differences in terrain height [25–27]. In terms of this article, these data form the foundation for building 3D models of buildings, as evidenced by the research of Loutfia [28] and Li [29]. However, the technologies used to create 3D models of buildings are burdened with long production times, which is a real obstacle to performing frequent updates.

Currently, there is a noticeable lack of systems that effectively integrate data from different georeferenced database resources, which is a significant challenge in the field of visualizing spatial objects in an AR environment. The integration of these data is crucial because spatial objects, such as buildings, infrastructure, or environmental features, need to be accurately mapped and situated in real space to enable accurate interaction with the real world. A system that provides for the integration of spatial data from georeferenced databases maintained by public administrations will allow for three important features:

1. **Data validity**—The data resources in the national spatial databases are updated relatively frequently and provide high-quality spatial data, especially in the case of orthophotos.

2. Data reliability—The data included in the national resources are produced according to specific standards, which guarantees the reliability of the data used to generate objects in the AR systems.
3. Data universality—Public administrations are obliged to make data universally accessible (requirements of the Infrastructure for Spatial Information in the European Community (INSPIRE) Directive) [30], which is implemented, for example, by carrying out photogrammetric measurements to produce an orthophotomap or by maintaining a vector database of buildings for the whole country.

The objective of the research presented in this paper is to design a multimodal system that integrates data from georeferenced spatial databases to visualize geographic objects on mobile devices using AR. This study focused on selected spatial objects representing buildings. The final result of the completed process is a solid representing the building objects along with the classified roof structure.

A crucial objective was to facilitate the integration of spatial data. The system employed both vector building outline data and an orthophotomap. The two aforementioned spatial data layers were incorporated into the system via the utilization of map services. In the case of the vector data, this was the WFS, while for the orthophotos it was the WCS (Web Coverage Service). This represents the initial component of the system tasked with the integration of spatial data. The key benefit of this solution is the assurance of the spatial data's validity, which is derived from georeferenced spatial databases that are updated on a regular basis. In the case of the building outlines, this occurs on a daily basis, while the orthophotomap is updated at intervals of one year or more. The data extracted using the aforementioned services provides the prepared building classification algorithm, which employs machine learning (ML) methods in YOLO 10. This constitutes the second component of the system, the purpose of which is to classify building objects into building blocks, representing them with their roof shape. The authors elected to utilize ML methodologies for the detection of roof structures, given the documented high performance of such approaches in prior research, as evidenced by the works of Castagno and Atkins [31], Rezaei et al. [32], and Campoverde et al. [33]. The third integral component is the visualization module of the obtained 3D models, which was implemented in an AR mobile application.

The system, in its final form, merely constitutes a preliminary phase in the development of a more streamlined and integrated multimodal system. The following key achievements have been made:

1. The integration of geospatial data with AR in real time represents a significant development in the field of geographic information science.
2. The utilization of deep learning algorithms (YOLO v10 XE) for the automated classification of roof types based on orthophotographs.
3. A proposal for future work includes the establishment of a scalable system for the management of international data.
4. The utilization of a semi-automatic approach, with the potential for progression to full automation, is imperative. This approach should be integrated with ALS LiDAR and oblique imagery.

2. Materials and Methods

2.1. Study Area

The geographical area selected for this study is situated in Poland, within the Lubelskie Voivodeship, encompassing the Lublin Poviát. The designated region encompasses a portion of the Głusk Community. This area is characterized by a relatively dispersed pattern of suburban development. The selection of this area is attributed to the substantial increase in building activity observed in recent years. The extent is shown in Figure 1. The

region that was selected is a representative area within the system that was executed. The system was designed in such a way that it is possible to adapt it to other regions through the use of open standards.

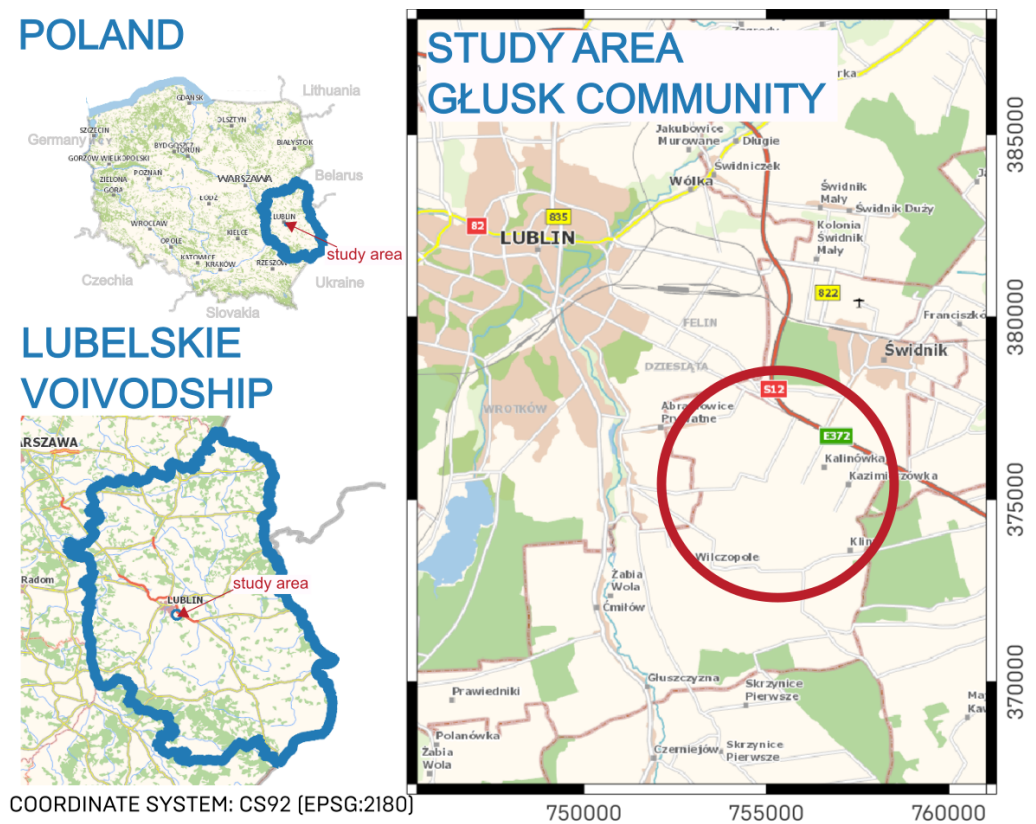


Figure 1. Study area in Głusk Community, Lublin Poviát, Lubelskie Voivodeship, Poland (source: author's own elaboration).

2.2. Data

This study utilized data made accessible as part of the national surveying and cartographic resource, encompassing orthophotos and comprehensive information on the geometry of buildings. In alignment with the openness policy adopted by the GUGiK, these data are publicly available and free of charge for acquisition [34].

The data may be accessed and extracted via the web-based National Geoportal application [35], which allows for straightforward access for both individual and institutional users. Alternatively, the data may be extracted via an automated method using web services that comply with international standards, such as WFS, which is designed to provide vector data, and WCS, which allows raster data to be downloaded. The utilization of open standards, such as WFS and WCS, endows the system with a substantial degree of flexibility, thereby facilitating seamless integration with a diverse array of spatial data sources, irrespective of geographical location or technological heterogeneity across different regions. The implementation of these open standards ensures compatibility with a comprehensive spectrum of geospatial databases, encompassing both national and international databases. This integration enhances the system's potential, facilitating easy adaptation to different regional specifications and requirements, which is crucial in a global context.

The geometric data of buildings included in the Land and Building Register database are recorded on the basis of land surveys. These measurements are taken along the contours of the buildings, at their point of contact with the ground. It should be noted that the dataset does not include information on the geometric roof details, which often extend beyond the building outline. The orthophotos used are studies with a high fixed spatial

resolution (pixel value of 0.05 m), providing precise spatial information about the actual land use. The use of an orthophotomap at this resolution facilitates the detection of roof details, including edges, chimneys, and skylights, with a high degree of precision. The georeferenced accuracy of such maps is generally 10–20 cm, enabling the precise rendering of roofs. The spatial data acquired are compiled in the CS92 system (EPSG: 2180).

It is imperative to underscore the limitations of orthophotos, which provide rudimentary information about roof geometry. Primarily, their two-dimensional nature hinders the accurate representation and recognition of complex roof forms. Secondly, the use of orthophotos as a reliable source for roof recognition is questionable. Future research should focus on integrating these data with more advanced sources to enhance their accuracy and utility. The integration of ALS LiDAR data will facilitate the creation of precise 3D models of roofs and enhance the discernment of intricacies. The utilization of oblique images will enable the observation of roofs from an alternative perspective, thereby supporting the precise identification of shapes, details, and proportions.

3. Results

3.1. Preparation of Test Data

The construction of the detection model constituted a pivotal component of the presented system. In order to achieve this, it was decided to utilize a conventional yet efficacious neural network model of the CNN (Convolutional Neural Network) type, such as YOLO (You Only Look Once). This model is distinguished by its rapid processing speed and high degree of accuracy, which renders it a valuable tool for the task of real-time object detection. Its single-shot architecture enables simultaneous detection and classification, significantly reducing data processing time and inference latency. Moreover, YOLO's lower memory requirements, compared with models like Faster R-CNN and EfficientDet, make it well suited for systems with constrained hardware resources.

The efficiency of YOLO extends to its transfer learning capabilities, where the single-pass approach simplifies the adaptation process to new datasets. This characteristic substantially shortens the time required for model fine-tuning, ensuring faster deployment in practical applications [36–39].

However, adapting the YOLO model to the intended classification necessitated a comprehensive fine-tuning process, which involved modifying the network parameters to align with the project's unique requirements.

In order to successfully perform fine-tuning, it was necessary to collect an appropriately sized dataset that would form the foundation for the subsequent training process. To this end, high-resolution orthophotos with a pixel size of 0.05 m were employed, providing detailed information about the objects under analysis. In order to minimize errors resulting from limitations in accuracy, a 2 m buffer zone was established around the contours of the buildings. This is intended to compensate for any discrepancies that may exist between the geometry of the buildings and the orthophoto data. Moreover, numerical descriptions of building contours extracted from publicly available resources on the National Geoportal were employed to augment the dataset. These georeferenced data constituted a precise starting material for model training, enabling effective classification and detection of spatial objects in the system. Thanks to such data, the YOLO model could be optimized to detect and classify different building types based on their geometric features.

The initial phase of this study entailed the delineation of distinct roof types through the manual identification of these features on the available orthophotos. It is anticipated that the utilization of orthophotos or quasi-orthophotos obtained from targeted unmanned aerial vehicle (UAV) flights will facilitate an enhancement in the temporal resolution of this study. Although the orthophotomap used in this study is a cartometric creation, the

photographed fragments of objects above the ground are not projected in an orthogonal manner. Hence, the previously described buffer zone of 2 m was introduced around the building outlines to account for these imperfections. Each building outline was then subjected to verification and subsequently assigned a corresponding roof type. The initial selection was subsequently divided into the following roof types: flat, shed, gable, hip, pyramid, and other roof types that could not be assigned to the aforementioned categories (Figure 2). It is worthy of note that in the sample under analysis, there were instances where the outlines of buildings did not align with the data presented on the orthophotos. This was most frequently the result of situations in which the contours indicated the presence of buildings that in fact no longer existed or those that had been rebuilt or developed. Such inconsistencies may be due to delays in updating databases, dynamic urban expansion, or differences in data acquisition methodologies, indicating the need for regular verification and integration of these sources of information. Consequently, these data have been omitted from further elaboration. A total of 3230 buildings were identified and manually assigned a roof type on the original orthophotos used.

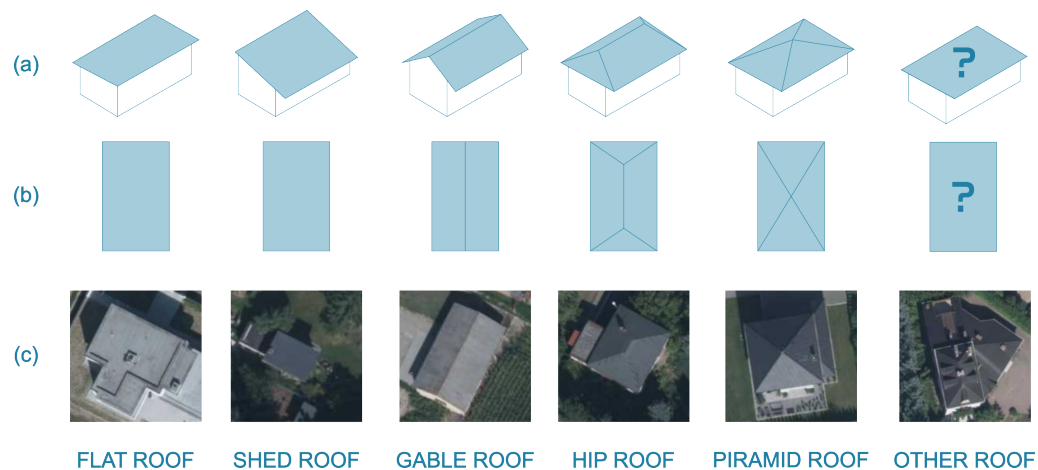


Figure 2. Repository of certain roof type classes: axonometric view (a) and top view (b) with geometrical parameters involving examples from orthophoto (c) (source: author's own elaboration).

3.1.1. Preparation of the Teaching Set for the CNN YOLO Model

In the subsequent phase, the automated preparation of the YOLO model training dataset was initiated and based on clipped orthophoto fragments, with dimensions of 2048×2048 pixels (equivalent to 102.4×102.4 m). These fragments were then subjected to annotation. The clipped fragments were created with an overlap of 200 pixels (10 m). This raster geometric resolution and overlap value were selected following an analysis of the physical size of the buildings and their architectural differentiation, as well as the orthophoto resolution. This approach ensured that each building with the assumed buffer fit in its entirety on the image clipped for training. The scaling process was conducted using bilinear interpolation, which guaranteed seamless transitions between pixel values and minimized the occurrence of scaling artifacts. A graphical representation of the aforementioned assumptions is provided in Figure 3. In accordance with the aforementioned methodology, 325 images were created. For each generated image, the requisite annotation text files were prepared in accordance with the specifications set out by YOLO. The format of the YOLO text file is as follows: <class_id> <x_center> <y_center> <width> <height>. The initial element is the object class (integer), followed by the coordinates of the centroid of the object, and then the dimensions of the object bounding box. The coordinates and dimensions are expressed in values from 0 to 1, defined within the local raster coordinate system, with the top left corner assigned the value (0,0).

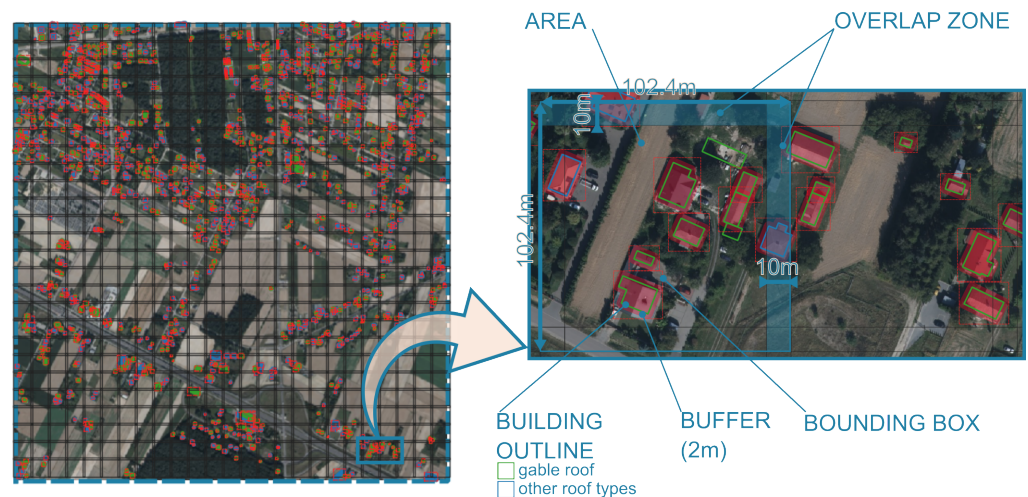


Figure 3. Scheme of assumptions for creating data for the learning set of YOLO (source: author's own elaboration).

Furthermore, chromatic normalization of the images was conducted to mitigate the influence of variable lighting and atmospheric conditions (present during image acquisition) on the quality of the input data. This normalization entailed the standardization of brightness and contrast values, which was pivotal for the stability of the ML models' execution and is typically employed in image data preprocessing.

3.1.2. The Selection of Representative Classes

The data prepared with the assignment of roof type classes to building contours were subjected to analysis. It was observed that gable roofs exhibited a notable prevalence in comparison with the other roof types. The graphical representation, which is a bar chart, is presented in Figure 4. The horizontal axis represents the numbers of the adopted areas (target fragments of the orthophotos), and the vertical axis represents the percentage share of each roof type class. Approximately half of the total number of roofs (with an average value of 51%) are gable roofs. The remaining roof types, in descending order of prevalence, are flat roofs (5%), shed roofs (10%), hip roofs (9%), pyramid roofs (10%), and other roofs (10%). These roof types collectively comprise the other half of the total classes. A normal (Gaussian) distribution graph is employed to present the percentage of classes (Figure 5). The flattened shape of the curve attributed to pitched roofs indicates a significant variation in the size of this class between areas. This class is present in each of the areas under analysis. In the majority of areas, the proportion is greater than 50%. In accordance with the three-sigma rule, in 99.73% of the areas under consideration, this class occurs in a proportion between 13% and 91%. The characteristic curves for the remaining classes are not as flat and are not consistently present in all areas. In comparison with gable roofs, the number of individual classes is significantly smaller.

Given the above interpretations of the graphical visualizations of the classes created, two representative classes were selected for further research. The division into two classes constitutes a preliminary step in the ongoing research project. Subsequent iterations of this study will involve the expansion of the number of classes to include a greater variety of roof types. The first one is gabled roofs, while the other classes were combined into one class as other roof types. These two created classes have a similar distribution in terms of their frequency in the created learning datasets. These classes were converted to YOLO in the resulting text files.

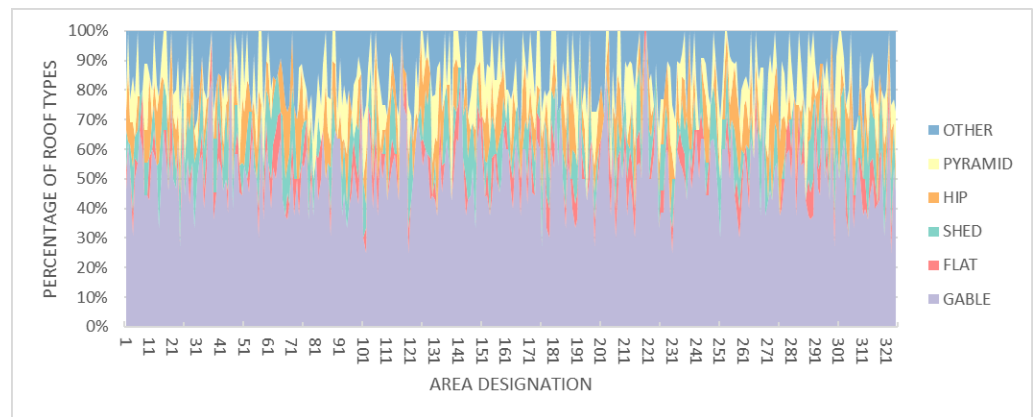


Figure 4. Diagram with the percentage of the roof type class in each individual area (source: author's own elaboration).

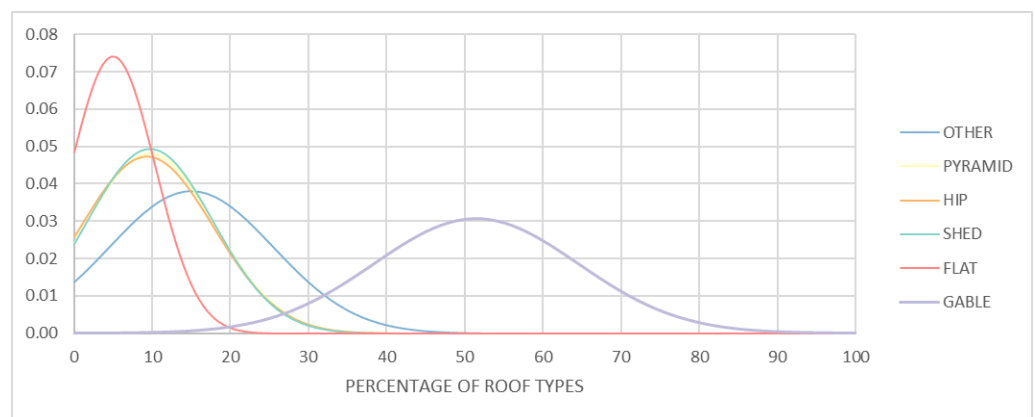


Figure 5. Normal (Gaussian) distribution graph of the percentage of roof type classes. (source: author's own elaboration).

3.1.3. Data Partitioning for CNN YOLO Model

The dataset generated for the CNN model was partitioned into training, validation, and test sets in the proportions of 70% (227 images), 15% (49 images), and 15% (49 images) (in accordance with generally accepted practices [40–43] in ML, allowing an effective assessment of the model's ability to generalize). Random partitioning of the data, with the granularity of the random generator controlled, provided repeatable results and an equal distribution of classes in each partitioned set, giving high confidence in avoiding class bias.

3.1.4. Patterns of the Training Data

After training the YOLO model, an analysis of the resulting graphical representation of the statistical data was carried out. A full description of the model's training process can be found in the following subsections and a discussion of the model's effectiveness is included in Appendix A.1. A full description of the training process and a discussion of the effectiveness of the model can be found in the following subchapters. The heat map of the label dimension (Figure 6a) shows no clear pattern or scheme in the distribution of buildings across the study areas. Buildings are evenly distributed across the study areas. An analysis of the dimensions of the bounding boxes (Figure 6b), which are defined as building outlines extended by a 2m buffer, shows that their dimensions are mostly between 0.10 and 0.15 for height and 0.10 and 0.20 for width, in line with the raster units used in the input data format for the YOLO model. In particular, a pattern of bounding box dimensions resembling a linear function was observed from the graphical visualization. A predominant number of objects have proportions approaching a square, which is characteristic of the

buildings analyzed in the dataset studied. These regularities may be evidence of the uniformity of object geometry in the training data, which may influence the effectiveness of YOLO model recognition in further applications.

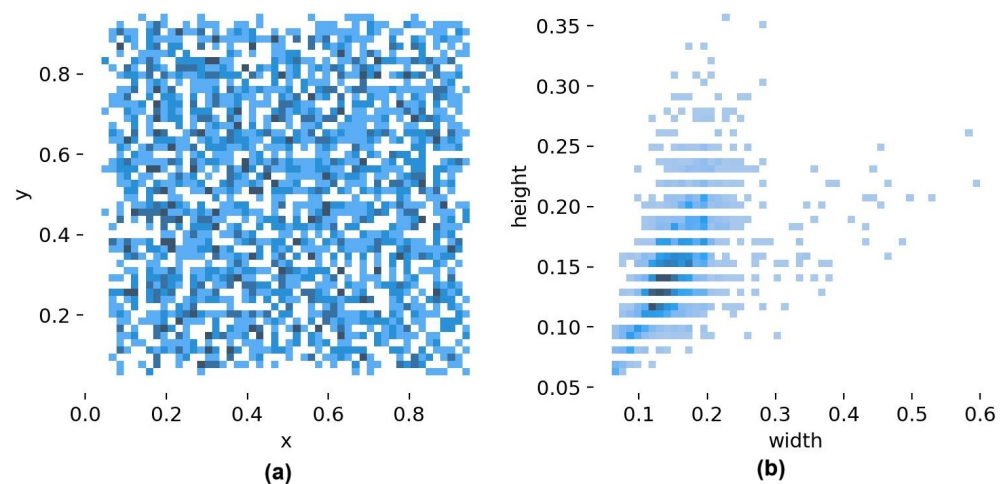


Figure 6. Statistics of training data from YOLO: (a) Centroid distribution of training objects. (b) Bounding box dimensions of training objects (source: author's own elaboration).

3.2. Evaluating the Efficiency of the YOLO v10 XE Model in Analyzing Building Roof Topologies

The project presented here employed algorithms for the analysis of geospatial data, utilizing techniques from computer vision, ML, and computational geometry. The YOLO v10 XE model for object detection on rasters was employed, which represents one of the most recent and sophisticated iterations of the You Only Look Once (YOLO) family of models. These models are renowned for their exceptional precision and efficiency in real-time object detection. It was anticipated that the YOLO v10 XE model would demonstrate high detection efficiency. A model performance evaluation was required to verify how effectively the system handled the classification of building roofs, taking into account their various typologies. The analysis included two main categories: gable roofs and other roof types, collectively referred to as 'all other'. This division permitted a more detailed assessment of the model's capacity to differentiate objects based on their shape and to identify potential issues associated with less typical roof constructions. To further enhance the system's capabilities, future work includes the integration of ALS LiDAR data and oblique imagery, aiming to achieve full automation of the roof classification process. This addition is expected to improve the model's ability to handle complex roof geometries and provide a richer dataset for analysis. This process necessitated meticulous data preparation, as previously mentioned in the preceding Section 3.1 *Preparation of test data*, rigorous training of the model, selection of appropriate evaluation metrics, and sophisticated tuning of hyperparameters were undertaken to optimize the classification results and ensure high generalizability of the model.

3.2.1. Modeling Parameter Calibration and Data Augmentation

In order to enhance the model's efficacy in identifying roofs on raster images, hyperparameter tuning and data augmentation techniques were employed. This was performed with the objective of augmenting the diversity of the training set and preventing overfitting, which is of particular significance given the limited amount of data. Hyperparameter tuning was conducted iteratively to refine the model's performance. The learning rate was initially set to 0.01 but was reduced to 0.001 after preliminary experiments showed that a higher learning rate led to unstable loss behavior in the early epochs. The weight decay

parameter, set to 0.0001, was used to regularize the model by limiting the magnitude of the weights, thereby reducing the risk of overfitting. These values were determined based on a combination of empirical tests and cross-validation on the training dataset.

The techniques employed were as follows:

Rotation: The images were rotated at random angles, thereby enabling the model to learn invariance to the orientation of objects.

Scaling: The size of the objects in the image was altered, thus facilitating the detection of objects of varying scales and aligning with the theory of scalability in neural networks.

Changing brightness and contrast: The model's immunity to environmental changeability was enhanced through the simulation of different lighting conditions.

As a result of image augmentation (each image in each epoch was randomly duplicated using 5 sets of augmentation types), the final result was $(1 + 5) \cdot 227 = 1362$ images in the training set. The augmentation of data was implemented exclusively on the training dataset, with the objective of ensuring an unbiased evaluation on the validation and test sets.

3.2.2. Training the YOLO v10 XE Model

The model was trained using the Adam optimizer, which is commonly used for its efficiency in convergence to minimize the loss function with large datasets and adaptive learning rate adaptation [44,45]. The training parameters, such as the batch size (set at 128) and the number of epochs, were selected based on experimental validation and learning curve analysis. The batch size had an effect on gradient stability and learning efficiency; the literature suggests that smaller batch sizes may lead to better generalization [46]. The number of epochs was increased to 350 with an early stop option (detected on the 327th epoch (Figure 7) in order to achieve full convergence of the model without the risk of overfitting. This was confirmed by the validation metrics, loss function analysis, and mAP (mean Average Precision). It is crucial to highlight that the model was developed through transfer learning, utilizing a pre-trained MS COCO (Microsoft Common Objects in Context) dataset. This approach involved freezing the weights of the backbone layers responsible for extracting general visual features, such as points, edges, and textures.

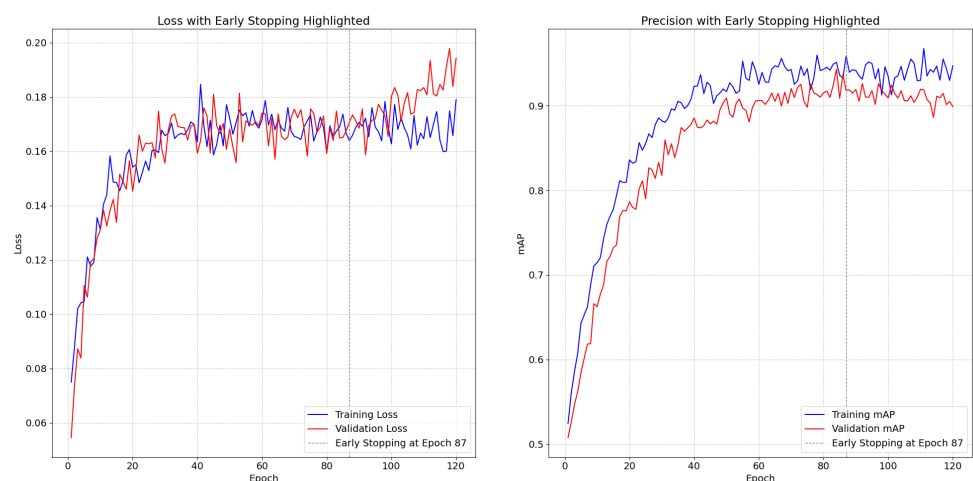


Figure 7. Training and validation loss and precision trends across 350 epochs. Early stopping at epoch 327 ensures optimal precision and prevents overfitting (source: author's own elaboration).

The evaluation of model performance was based on a number of metrics, including precision, recall, the F1-score measure and accuracy. These metrics are standard in the evaluation of classification models and facilitate a comprehensive analysis of model performance [47–49]. A detailed description of the indicators used is included in Appendix A.1.

Using training data for only two classes in the YOLO v10 XE model and limiting the area to one region results in a simplified and unbalanced set. During data preparation, it was found that one class (gabled roofs) had a significant advantage over the others. It was therefore decided to merge the other classes into one (other roofs). This may result in an imperfect classification for this second class. The YOLO model may focus on the dominant class and ignore the less common roof types, resulting in low classification accuracy. The presence of irregular roofs constituted an additional challenge. The presence of objects of this nature resulted in a generation of false positives and false negatives during the classification process. The background complexity of the image data also proved to be a significant difficulty. The variety of backgrounds (e.g., the presence of dense vegetation, shadows, vehicles, and other infrastructure elements) affected the stability of the model during the optimization process. The model encountered difficulties in distinguishing target objects from background noise. It was also evident that there was an issue with the convergence of the model during the training process, particularly in the initial epochs. The extensive number of parameters in YOLO v10 XE necessitated meticulous calibration of the hyperparameters to circumvent substantial fluctuations in the loss function values. Consequently, the training process was susceptible to the phenomenon of overfitting to more prevalent examples in the data, while disregarding more challenging instances.

3.2.3. Analysis of Misclassified Examples

Despite the high accuracy achieved by the YOLO model in detecting the two classes of roofs, a small number of identification mistakes remain, attributable to the complexity of roof shapes and the variety of objects that can be misclassified as roofs. One of the most common errors is the misclassification of a roof, especially in the case of unusual roof forms that differ from traditional shapes, such as roofs with complex geometry (e.g., roofs with additional superstructures) (Figure 8a). Additionally, the presence of supplementary objects that may be constructed from a similar material to the roof (e.g., sheet metal, shingles) can result in erroneous recognition by the model. The development under discussion is also influenced by highly compact development, where the roofs of individual buildings lose their extent (e.g., one roof for several buildings).

Another factor that impacts the quality of the classification is the presence of objects that have a roof covering but do not constitute buildings, such as sheds, gazebos, or other small structures (Figure 8b). The model may misclassify these objects as part of a larger building structure because their roofing may resemble a classic roof shape. Additionally, there are instances where objects lacking a roof may be erroneously classified as such, particularly when they exhibit a rectangular form with a distinct delineation, which may imply the presence of a roof in the analyzed area (Figure 8c).

Furthermore, there is a paucity of recognition of certain roof types. This is primarily due to the unconventional configuration of the roof, which significantly deviates from conventional designs (e.g., conical roofs or roofs with numerous additional elements) (Figure 8d). Errors of this nature are particularly evident in instances where the roofing is diminutive in size, occurs on an unconventional structure, or is composed of materials similar to those utilized on building roofs. The development of a multimodal system that will make use of additional data, such as ALS LiDAR or oblique images, will necessitate the improvement of the YOLO model to eliminate these errors. Moving forward, there is a need to focus on enhancing the capacity to distinguish objects not only based on their geometry but also considering additional features such as height, slope angle, and material details. This will help to minimize misclassification. With further development and integration of diverse data sources, the system will be capable of more accurately identifying roofs and differentiating between objects with similar structures.

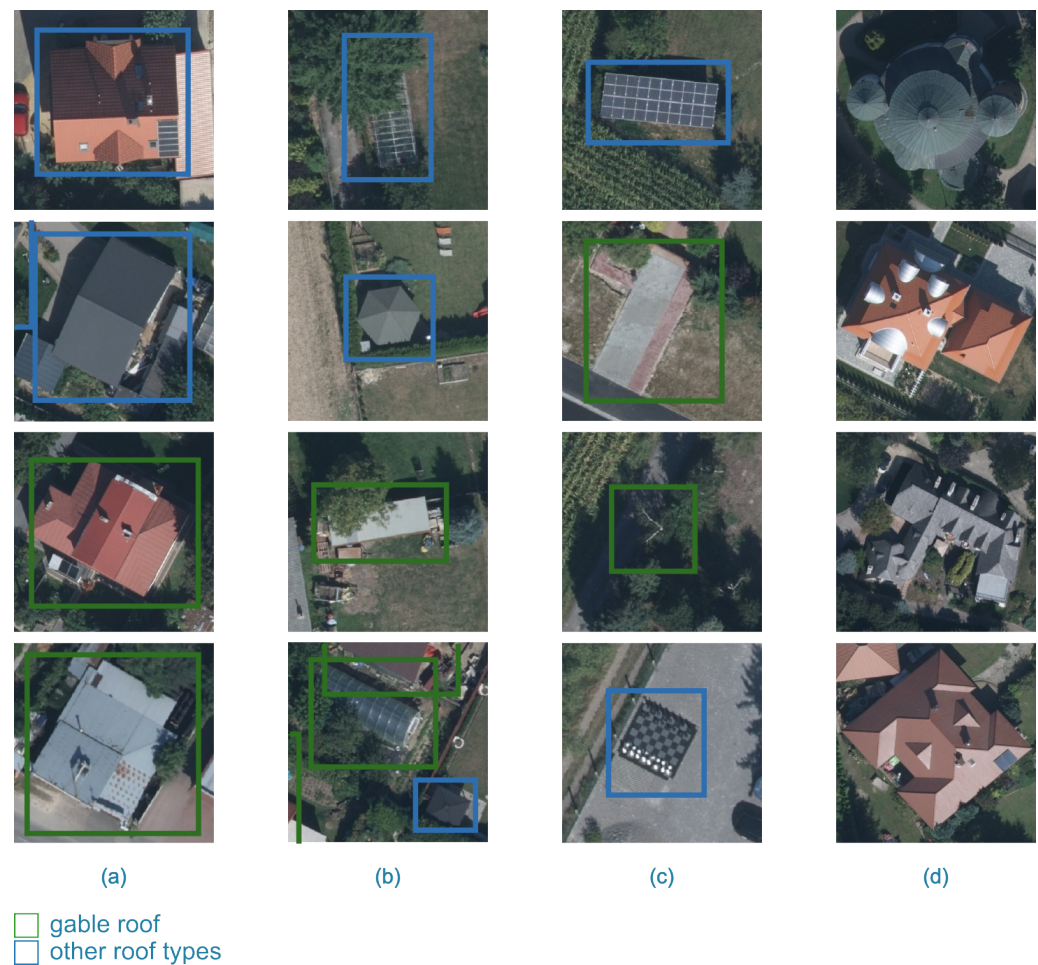


Figure 8. Examples of misclassification of roof types: (a) incorrect class, (b) roof is not located on the building, (c) incorrect object other than the roof, (d) unclassified (source: author’s own elaboration).

3.2.4. ANOVA (Analysis of Variance)

To further elucidate the efficacy of the YOLO model, an experiment based on an analysis of variance (ANOVA) was devised. This analysis assesses whether there are statistically significant differences in the effectiveness of the YOLO model between the two types of house roofs detected in the orthophoto images. Two classes of objects were included in this study: houses with gable roofs and others. The performance of the model was assessed using three metrics: precision, recall, and F1-score. The F1-score is a measure of the ability of the model to detect each class but realized for each subarea.

The formulas and results of the ANOVA analysis are described in detail in Appendix A.2. The results indicate that the differences between roof types are on the verge of statistical significance ($p = 0.07$), which is likely due to the dominance of gable roofs in the training dataset and the limited representation of other roof types. These factors emphasize the need for a more balanced and diversified dataset to improve the robustness of the analysis.

The ANOVA analysis conducted allowed a detailed evaluation of the YOLO model, revealing differences in detection performance between the two roof types. These results suggest the need for further tuning of the model and the need to supplement the training data with examples of other roof types. Marginalizing these differences would produce more consistent results and improve the overall performance of the model.

3.2.5. Procedure for Conducting the Experiment

To ascertain the repeatability and reliability of the results, multiple analyses were conducted on different test areas. Precision, recall, and F1-score values were calculated for

each area, thus enabling the efficiency of the model to be evaluated under varying building layouts and environmental conditions.

The procedure entailed the following steps:

- A variety of samples were tested, with the selection of areas featuring different architectural styles and building densities allowing for the results to be tested universally and avoiding the errors associated with a homogeneous test set.
- Testing on a variety of samples—Selecting areas with different architecture and building densities allowed the results to be tested universally and avoided the errors associated with a homogeneous test set.
- Statistical Analyses—Statistical calculations such as mean precision and standard deviation were used to determine the stability of the model and the level of variability of the results in the different test areas.

3.3. Create Interactive 3D Representations of Building Roofs

The basic concept of the multimodal system created was to visualize the buildings in 3D, taking into account both their solids and the shapes of their roofs. The project developed two 3D representations (Figure 9) corresponding to the two classes of roof types distinguished: gabled roofs and other types. Such an approach has made it possible to effectively represent the differences in the geometry of the buildings, while at the same time allowing a spatial analysis of their architectural characteristics in a more visually intuitive and accessible form.

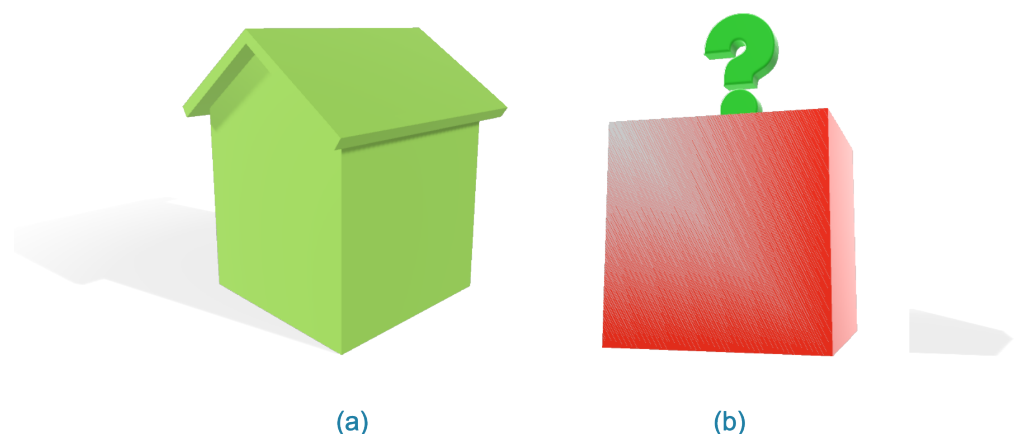


Figure 9. Visualization of representative classes of building blocks: buildings with gable roofs (a) and others (b) (source: author's own elaboration).

To create representative solids for the two selected roof classes, we used the free software Blender <https://www.blender.org/>, which is becoming increasingly popular in fields such as architecture, urban planning, and the development of geographic data in the context of GIS [5,6]. Thanks to its universality, Blender provides a tool that allows not only 3D modeling but also integration with a wide range of spatial applications. The 3D building models created in this program were exported in glTF (Graphics Library Transmission Format), which allowed them to be implemented in the three.js library, a popular tool for rendering 3D graphics in a web application environment. This approach allows models to be dynamically presented in web applications and easily integrated into interactive GIS systems.

3.4. System Structure and Data Implementation Processes

The development of a web application using WebXR technologies and the three.js library provides a dynamic exploration of 3D building models in an AR environment. The tools used allow users to interactively visualize building solids, combining an immersive

AR experience with an accurate representation of spatial data. This paper discusses both the technical assumptions of the project and the key implementation aspects related to the integration of geospatial information and spatial visualization.

3.5. Development of the WebXR Application

The web application was developed using the following tools: the three.js library [50] and the WebXR API [51]. The three.js library was chosen to generate 3D animations based on pre-designed representations of building solids. These animations are designed to allow users to dynamically explore models of buildings, including their solids and roofs. The animations generate the following effects: rotation and perspective change, zooming in and out, interactive navigation, and AR usage. This enables interactive 3D models to be rendered and displayed in web browsers, allowing users to accurately view buildings in virtual space. The WebXR API has been used to integrate with AR and VR technologies, allowing users to interact with the 3D model in a more immersive way, both on mobile devices and using VR goggles. This library allows previously created data to be visualized in 3D and displayed in the application. The building solids are displayed according to the coordinates of the building centroids. Each model is loaded with the same dimensions and orientation, regardless of the building's outline and location. The selection of the appropriate representative solid is based on the class assigned to the building. The building class and centroids are determined from the object recognition performed on the orthophoto.

3.6. Mobile User GNSS Data

The developed application works in the local Polish CS92 system (EPSG: 2180), based on the GRS80 ellipsoid. The source data extracted from the National Geoportal [35] are compiled and made available in this system. One of the objectives of the multimodal system was to automatically acquire or manually assign the location of the mobile user's device. The geographical coordinates extracted from the mobile device are in the global WGS1984 system (EPSG:4326). The coordinates are transformed into the cartographic coordinate system in which the application is operating. This is achieved through the use of the Proj4js library [52], which employs a mathematical transformation utilizing Gauss–Kruger projection (a variation of Merkator transverse projection). This transformation enables the conversion of geographic coordinates into flat orthogonal coordinates, taking into account the parameters of the destination system, such as the central meridian or offset and scale. The library automatically implements this method using defined EPSG standards, thereby ensuring compliance with accepted geodetic specifications. The final coordinates are used to determine the centroid of the location of the extracted orthophoto section for visualization using the WCS service. The coordinate range of the extracted area is set by the user, using the dimensions of the raster (width and height).

System Architecture and Data Flow

Figure 10 illustrates the general concept of data flow between the principal components of the system. The mobile user utilizes a browser-based mobile application, which enables the device's location coordinates Global Navigation Satellite System (GNSS) to be obtained and transmitted to another component of the system (the Django application server) via HTTPS (HyperText Transfer Protocol Secure). Additionally, the browser receives results from the system, including building perimeters and data for visualization in geojson format. In this context, Django is the primary data processing component. The application server acquires the coordinates from the browser and then communicates with the database, geospatial services (downloading orthophotos from the National Geoportal), and the classification module.

The Django framework is responsible for the following processes:

- In order to ascertain the veracity of the data in question, Django verifies the local spatial data resources. In the event that the user's location is known, the information is not obtained from external services, but rather from the cache of the aforementioned resources.
- The processed results are stored in a PostgreSQL database for potential future use in subsequent location-recursive requests.
- The system generates building boundaries based on data collected or downloaded from external services.
- Django stores information about the roof types detected in the analysis for a given user query area.
- Django sends queries to external WFS and WCS services to retrieve geospatial data for a given area.
- Download from an external source of building boundaries and orthophoto data.
- The YOLO model is used to initiate roof detection and classification tasks on selected areas of the orthophotos, in order to assign appropriate building models to the visualized space in AR.

PostGIS (PostgreSQL): Acts as a geospatial database, storing local information about buildings, roof types, and analysis results. Django communicates with PostGIS to read and write spatial data.

National Geoportal [35]: An external source of geospatial data from which Django extracts building boundaries and orthophotos. WFS and WCS services are used to extract current data.

YOLO v10 XE: A classification module that analyzes images to detect and classify roof types based on orthophotos.

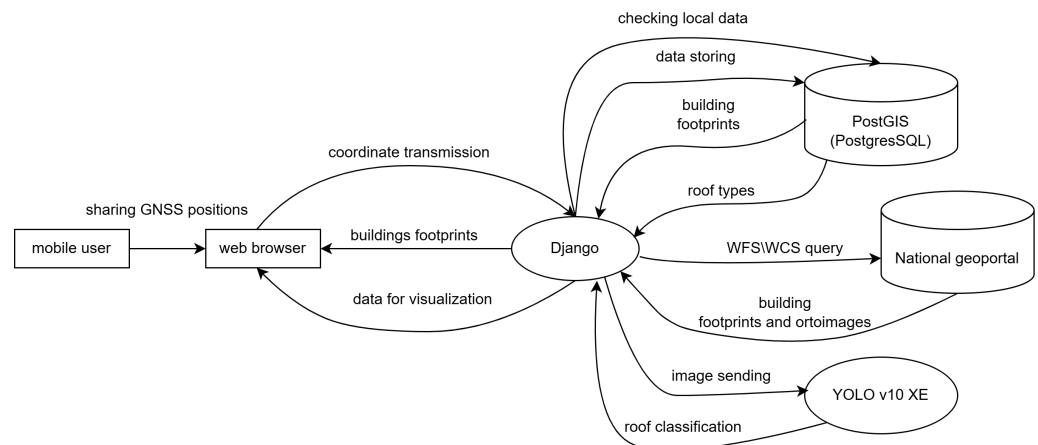


Figure 10. Multimodal system component data flow architecture (source: author's own elaboration).

4. Advantages of Visualizing Results in AR in a Mobile Application

The classification of roof types in a defined area is very useful in aspects related to architecture, land use planning and management [53], real estate (including valuation) [54], environmental engineering [55]. An important aspect for understanding and interpreting the results is their visualization (Figure 11). Presenting them as a two-dimensional (2D) visualization through spatial distribution diagrams (scatter plot, heat map, contour plot) or as a map helps to understand the spatial structure of a place (city, district, street). This is made very easy to interpret and understand. One can see the patterns that occur in the location of buildings with a particular type of roof, where they are more or less concentrated. The resulting visualizations can suggest or directly depict different architectural styles, the function of buildings, their purpose, when they were built, and how they relate to

their location and neighborhood. Many of these visualizations reveal local architectural and urban trends that shape the character of urban areas. Traditional visualizations based on two-dimensional projections result in the loss of valuable topological information, especially in relation to height aspects. In contrast, 3D visualizations—including accurate representation of roof shapes, dimensions, and heights—tend to provide a more complete and objective picture of the project [16,32,33,54].

The use of VR technology significantly improves the quality and effectiveness of architectural visualizations [56]. AR allows buildings to be displayed in their actual location so that they can be viewed on a 1:1 scale in the context of the actual site conditions. This technology allows users to easily see how existing and designed objects will fit into existing buildings, improving design decision-making. AR also supports better spatial understanding, making it easier to identify potential problems with visibility, proportions, or how new buildings will blend in with their surroundings. In addition, AR can be used to interactively view models in real time, allowing for immediate design modifications and better alignment with user expectations [57].

The introduction of AR technology into design processes not only improves the aesthetic perception of visualization but also significantly enhances its functionality, engaging users in a more intuitive and interactive way. This makes it possible to more accurately predict the impact of new buildings on urban space and better plan the development of built areas.

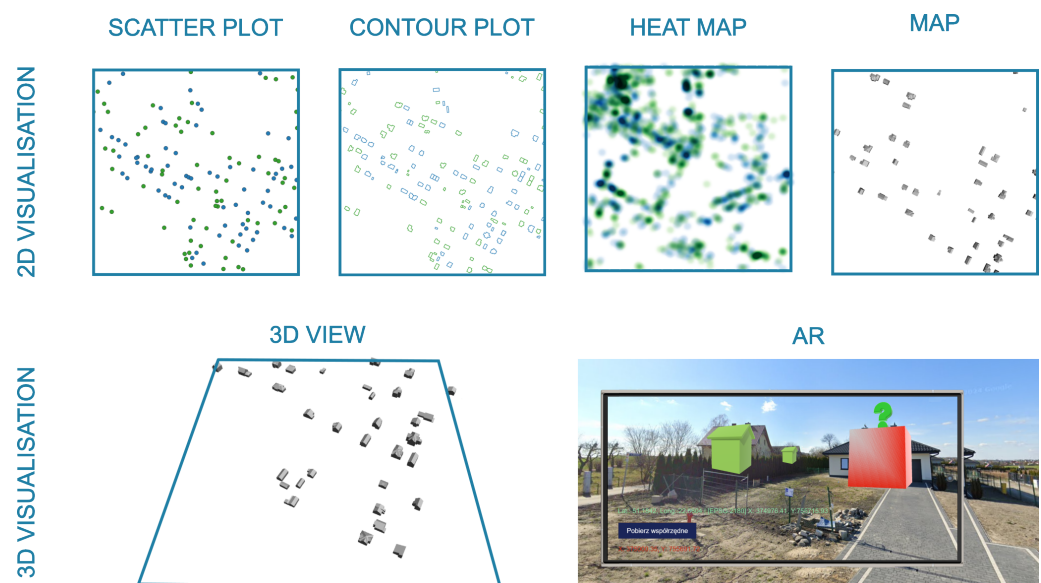


Figure 11. The options for presenting classes of building roof shapes in 2D and 3D (source: author's own elaboration).

5. Discussion

The research carried out is the first step in the creation of a universal multimodal system. The aim was to assess the feasibility of classifying roof types using YOLO. The system created is still incomplete and requires further development. Directions for the development and improvement of the system are discussed in the following section.

5.1. Model Limitations and Future Directions

The main objectives of the development of the system will be to expand the datasets to include more complex scenarios, balance the classes, and optimize the YOLO model.

In consideration of the coordinate range of the cropped area, the system may be enhanced through the adaptation of this area according to a trade-off between the user's

visual perception and the technical requirements of the image resolution employed by the ML algorithms. In order to define the optimal area for analysis, it is necessary to take into account two factors. Firstly, the human eye's ability to recognize detail must be considered, and secondly, the data processing capabilities of the mobile device must be taken into account. This includes its computational performance and memory limitations. Furthermore, mechanisms to automatically scale the analysis area according to the changing quality of the input data and available computing resources should be considered. This adaptive approach to determining the coordinate range of the analyzed area has the potential to enhance both recognition quality and application response time, a crucial consideration in the context of mobile applications with limited computing resources. The size of the analyzed area in the executed multimodal system does not affect its efficiency. The calculations and the data acquired are transferred to a server—scalability through server-side processing. This provides more computational capacity than a mobile device and the possibility for multiple users to operate simultaneously.

5.1.1. Classification Extension

The integration of additional roof types (new classes) into the model has been shown to enhance the system's versatility, thereby facilitating more precise identification of diverse structural forms. The augmentation of classes within the YOLO model is expected to result in a more comprehensive representation of roof typologies, encompassing less prevalent or more intricate shapes that are prevalent in real-world settings. This development is anticipated to enable more precise classification of roofs across diverse geographical regions, where a confluence of distinct building traditions and construction technologies is observed. The outcome of this research is expected to be a system that is both more flexible and resistant to errors associated with overlooking less typical constructions. This, in turn, should contribute to the system's widespread use in different geographical and industry contexts.

Augmenting the number of classes in the YOLO model will also have a bearing on the quality of the visualization generated in AR and VR, as it will allow for a more accurate representation of the real world. By classifying roofs with greater precision, the representation of these objects in VR and AR will become more realistic (higher accuracy), thus increasing their utility.

A significant element to consider is the incorporation of class types with associated elements and attributes. The optimal solution would be the creation of distinct classes for each additional element. Of particular importance is the material from which the roof is constructed, as this attribute will have a significant impact on the process of checking the roof for hazardous materials, such as asbestos [58]. An important aspect is the elements that cover the roof (e.g., solar panels, roof windows), the elements that modify the basic type of roof shape and these are most often the elements that light up the upper floors and attics. In most cases, roof windows (skylight) or rooflights are used, which do not affect the shape of the roof but the roof covering. Shape changes are influenced by the form of the dormer window or the so-called 'bull's eye'. The latter can significantly affect the recognition of roof types from orthophoto data. The construction of the roof and its basic shape can differ significantly from the photographic interpretation, the quality of which also affects the effectiveness of the classification. Most of the available and produced photographic map studies are classical orthophotos (objects close to the surface are rectified to an orthogonal position). This leads in some cases to a misrepresentation of the roof shape and a lack of coverage of the building outline [59].

5.1.2. Misclassified Objects

The identification of objects within the system is not without error. There is a paucity of misclassifications of objects. These are a result of the complexity and originality of individual cases, which differ significantly from objects in the environment. They are influenced by unusual roof forms, elements located on roofs, and the high density of buildings. In order to resolve this issue, it would be beneficial to augment the number of objects in the prepared data for the YOLO model and to establish an additional class to accommodate the misidentified objects.

Advanced segmentation methods are planned to improve the effect of overlapping objects on densities. The use of semantic segmentation through integration with models such as FCN (Fully Convolutional Networks) or instance segmentation (Mask R-CNN method) can improve the results.

5.1.3. Impact of Input Quality on Model Performance

The resolution of the input images is critical to the accuracy of object detection. Too low a resolution can lead to a decrease in object detection accuracy in the YOLO model. Conversely, too high a resolution increases computational requirements and can lead to model overfitting [60]. The quality of the data, including their completeness, consistency, and freedom from errors, has a direct impact on the performance of the model. The original resolution of the input images is critical for efficiency, especially when distinguishing fine details such as roof breaklines or roof texture.

5.1.4. Generalization of the Results

To increase the accuracy of the system and its capability, the accuracy of the input data for prediction training would need to be expanded and improved. A true orthophoto (all objects are rectified to an orthogonal projection) view of the building shape would be used, which is more accurate and corresponds to the building outline lines. The data could be supplemented with oblique images and ALS LiDAR data.

To improve the robustness of the model to a variety of conditions and to increase its generalizability, data augmentation techniques (e.g., rotations, scaling, changes in brightness or contrast, offsets) [22] were employed in this study to increase the size and diversity of the training set. These methods helped the model to generalize better by simulating different scenarios. In future studies, further improvements could be achieved by incorporating additional augmentation strategies and more diverse input data. For example, utilizing images captured at different times of day and in different weather conditions will increase the diversity due to changing lighting and different roof appearances (e.g., dry and wet roof textures). Including data with these variables in the training set will improve the model's ability to identify data in different circumstances.

At present, the system has only been developed for one region, which significantly limits its versatility. Expanding the training set with data from different regions will increase the variety of roof shapes and their textures. The use of data from different sources in the future development of the multimodal system will be crucial for its universality. Data from different sensors (aerial imagery, satellite imagery, and drone data) can support a better understanding of the specifics of different roof types. Future plans include the use of data from OpenStreetMap (OSM), GoogleEarth, and INSPIRE. The next phase of system development will be the use of synthetic data. ALS LiDAR data will be used to create 3D roof models and render them under different lighting and weather conditions.

Taking the above into account will allow a better YOLO model to be created, thus improving the multimodal system. Data classification will be able to handle a greater variety of data in practical applications.

5.2. The Purpose of the System

This system is the basis for utilization in urban planning, civil engineering, and emergency management. It involves a more accurate perception of the real world, supported by virtual worlds and visualizations. The identification and 3D modeling of building solids, including roof shapes, is important for urban planners, architects, and heritage professionals. The additional attributes that can be added to the system will have a significant impact on crisis management, including environmental management.

AR techniques used in the system have been shown to have a significant impact on the field of urban design and architects, primarily by enhancing community participation in the decision-making process, promoting a more comprehensive understanding of the subject matter, and enabling interactive feedback [17–21]. These techniques assist designers in evaluating and modifying ideas based on community input, thereby leading to more accurate design decisions. Furthermore, AR has been demonstrated to contribute to the enhancement of a city's image by showcasing both future and historical content, thus making it a valuable tool in urban development projects that emphasize the city's identity. In addition to this, AR plays a pivotal role in utilizing user experience to improve urban sensing and data collection. The effectiveness of AR in urban projects is contingent on the availability of technologies, expertise, and the scale of the project.

5.3. Future of the System

The potential of nature-inspired methodologies to enhance image-based roof classification is a promising avenue for future research. The integration of such approaches holds significant promise for expanding the application of the YOLO framework beyond its current scope, encompassing more intricate systems. The utilization of nature-inspired frameworks, exemplified by the BioVRbot toolkit [61], emphasizes the optimization of recognition and classification methodologies. The integration of these techniques has the potential to enhance the capabilities of the YOLO detection algorithm, thereby facilitating enhanced recognition of roofs characterized by challenging geometries or those that are obscured. Adaptive control systems, exemplified by H ∞ control in adaptive cruise control systems, can enable dynamic adjustments to detection parameters. This approach would be of particular utility in the management of large datasets, where high accuracy is required over extensive areas. A further enhancement could be to extend the system to incorporate robotics. Utilizing drones or autonomous vehicles equipped with a real-time rooftop classification system [62–64] would facilitate autonomous mapping and classification of roof shapes in conjunction with YOLO.

6. Conclusions

This paper's focus was on the development of a system that facilitates the effective integration of geophysical perimeter data with AR. A deep learning method was devised for the automatic detection of roof types from image data, specifically orthophotos. This study incorporated two roof types: gable roofs and other roof types. This enables a realistic representation of buildings and their visualization in three dimensions (3D). The results of the analyses conducted indicate the borderline statistical significance of the differences in the effectiveness of the YOLO model in classifying roof types. The metrics of precision, sensitivity, and F1-score yielded higher results for gable roofs (0.79, 0.78, and 0.78, respectively) in comparison with the other roof types (0.76, 0.75, and 0.74). The findings suggest that the model exhibits a higher degree of efficiency in recognizing objects characterized by regular geometry. The ANOVA results obtained indicate the need for further optimization of the model in the context of roof differentiation. The aim was to establish this empirically. It is recommended that the model be further refined and that the

training dataset be expanded to encompass additional examples of roof types, with a view to enhancing the consistency and accuracy of the results.

The integration of interactive 3D visualizations with AR has been shown to enhance the clarity and quality of the presentation of results to a significant degree. The utilization of libraries such as three.js and the WebXR API facilitates the dynamic exploration of building models within an AR environment. The adoption of 3D visualization techniques facilitates a more profound comprehension of the geometric disparities among edifices with disparate roof types, thereby enabling users to perform spatial analysis in an intuitive manner. The integration of such technological advancements into design processes fosters enhanced comprehension of spatial content contexts, thereby facilitating optimized architectural decision-making.

Despite the advantages of the developed system, its current classification of roof types is limited to two categories. The expansion of this classification to encompass additional roof types, as well as elements such as roofing materials, roof windows, and dormers, has the potential to enhance its usability significantly. The integration of data from object-based orthophotos, oblique imagery, and ALS LiDAR could lead to improvements in the accuracy of the model and the quality of visualization [65–68]. ALS LiDAR data facilitate the accurate mapping of roof heights and structures, thereby enhancing the model's capacity to differentiate between roof types with complex forms. The oblique imagery, with its varied viewing angles, provides a superior representation of roof fragments that are challenging to access and may be overlooked in conventional orthophotos. The integrity of these data sources enables the automatic generation of precise roof shapes within the system, thereby facilitating a more automated process. The integration of ALS LiDAR and oblique imagery serves to eliminate errors arising from the utilization of individual data sources, thereby enhancing the precision of roof detection and classification. Consequently, the development of a multimodal system will contribute to enhancing the accuracy and efficiency of roof recognition in diverse application contexts. The enhancement of such a system could find applications in fields as diverse as urban planning, disaster management, environmental engineering, and property management. The developed system has the potential for wide application in the fields of urban planning, architecture, and heritage management. The possibility of 3D modeling, in conjunction with AR technology, facilitates a more comprehensive spatial analysis and provides decision support for urban design and planning. The incorporation of additional attributes and enhanced classification algorithms would enable a more accurate representation of spatial structures and their functions, which could support emergency planning and environmental engineering. Subsequent endeavors will encompass usability studies and consultation with end-users, with the objective of enhancing the practical value of AR visualizations in architectural and urban planning processes.

The development of the application should encompass the expansion of the classification of roof types and the incorporation of additional elements (e.g., solar panels, roof windows) and attributes (e.g., roof material) related to roofs. Extensibility to differentiated datasets and adaptation to different geographical standards is supported by the system's architecture. The integration of data should be extended to encompass additional data sources (e.g., phone images, scanning data), and the application itself should be compatible with the infrastructure for INSPIRE, the national one, and easily applied to European infrastructure. This will result in greater versatility and unification of the geometric data and object description information created.

The present study constitutes a preliminary investigation into the capabilities of the YOLO model within a defined context. Subsequent research will concentrate on more sophisticated AI algorithms and intricate datasets. The proposed approach will entail the

optimization of classification in areas of high building density. This will be achieved by implementing advanced segmentation and input filtering techniques. The objective of the present study is to conduct a series of experiments on alternative architects with a view to enhancing performance in relation to irregular roof types.

The relocation of computing to the server side is a strategy that has been proven to ensure the scalability of the system. In addition, it has been demonstrated that this approach facilitates the efficient processing of large volumes of data, while concurrently imposing minimal load on mobile devices. Consequently, this methodological approach enables the system to be utilized extensively in real-world settings. The implementation of such a solution will enable the system to expand through the integration of sophisticated methodologies from diverse academic domains, thereby augmenting the scope of the available data. This will result in a substantial enhancement of the system's practical impact and its applications within dynamic urban environments.

Author Contributions: Conceptualization, B.K., A.J. and A.I.; methodology, B.K., A.J. and A.I.; software, B.K., A.J. and A.I.; validation, B.K., A.J. and A.I.; formal analysis, B.K., A.J. and A.I.; investigation, B.K., A.J. and A.I.; resources, B.K., A.J. and A.I.; data curation, B.K., A.J. and A.I.; writing—original draft preparation, B.K., A.J. and A.I.; writing—review and editing, B.K., A.J. and A.I.; visualization, B.K., A.J. and A.I.; supervision, B.K., A.J. and A.I.; project administration, B.K., A.J. and A.I.; funding acquisition, B.K., A.J. and A.I. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

Acknowledgments: Financial support of these studies from Gdańsk University of Technology from the DEC-33/1/2023/IDUB/I3b/Ag grant under the Argentum Triggering Research Grants—Excellence Initiative—Research University' program is gratefully acknowledged (A.I.).

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

3D	Three-Dimensional
ALS	Airborne Laser Scanning
ANOVA	Analysis of Variance
AR	Augmented Reality
CNN	Convolutional Neural Network
EPSC	European Petroleum Survey Group
FCN	Fully Convolutional Networks
GIS	Geographical information systems
glTF	Graphics Library Transmission Format
GNSS	Global Navigation Satellite System
GUGiK	Head Office of Geodesy and Cartography
HTTP	Internet's Hypertext Transfer Protocol
INSPIRE	Infrastructure for Spatial Information in the European Community
LiDAR	Light Detection and Ranging
mAP	mean Average Precision
ML	machine learning
MS COCO	Microsoft Common Objects in Context

OSM	OpenStreetMap
VR	virtual reality
WCS	Web Coverage Service
WFS	Web Feature Service
WMS	Web Map Service
YOLO	You Only Look Once

Appendix A

Appendix A.1. Evaluation and Assessment Metrics

The terms “true positives” (TP) and “false negatives” (FN) refer to the number of events belonging to a specific class that were correctly and incorrectly classified, respectively. “True negatives” (TN) represent the number of events that were correctly identified as not belonging to a particular class. Finally, “false positives” (FP) indicate the number of events that were incorrectly categorized as belonging to a specific category.

The accuracy metric represents the overall effectiveness of the model. However, it can be misleading in the case of unbalanced datasets. Therefore, the additional metrics provided below should be used in conjunction with the accuracy metric.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (A1)$$

Precision is a measure of the model’s capacity to avoid false positive classifications, which is of paramount importance in the context of object detection, where misidentification can have significant and detrimental consequences.

$$Precision = \frac{TP}{TP + FP} \quad (A2)$$

The recall represents the model’s capacity to identify all instances of true positive cases.

$$Recall = \frac{TP}{TP + FN} \quad (A3)$$

The F1-score represents the harmonic mean of precision and sensitivity, thereby facilitating a balanced assessment of the model.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (A4)$$

The resulting model achieved a high level of precision, with a value of 0.79, as determined by plotting the precision, recall, and F1-score metrics (Figure A1) for confidence threshold = 0.85. This is in agreement with the results of other studies in the field of building roof detection using deep neural networks. The low standard deviation between the different test areas is indicative of the stability and generalizability of the model.

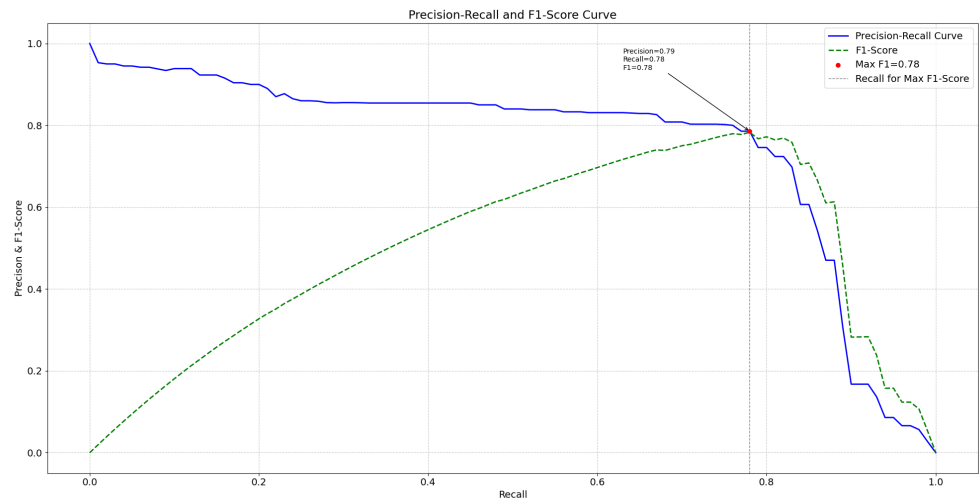


Figure A1. Determining the optimal F1-score value for confidence threshold optimization, considering overfitting (source: author's own elaboration).

Appendix A.2. Results of the ANOVA Analysis

The F statistic in the analysis of variance (ANOVA) was calculated according to the equation:

$$F = \frac{MS_{between}}{MS_{within}} \quad (A5)$$

where the following are used:

- $MS_{between}$: Mean squares between groups—a measure of the variance between groups.

$$MS_{between} = \frac{SS_{between}}{k - 1} \quad (A6)$$

- MS_{within} : Mean squares within groups—a measure of variance within groups.

$$MS_{within} = \frac{SS_{within}}{N - k} \quad (A7)$$

- $SS_{between}$: Sum of squares between groups—measures the difference between each group's average and the global average.

$$SS_{between} = \sum_{i=1}^k n_i (\bar{X}_i - \bar{X})^2 \quad (A8)$$

- SS_{within} : Sum of squares within groups—measures the difference between individual observations and the group mean.

$$SS_{within} = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2 \quad (A9)$$

where the following are used:

- x_{ij} is the value of observation j in group i .
- $\bar{X}_i$ is average for group i .

- k is the number of groups.
- n_i is the number of observations in group i .

Table A1. The results of the ANOVA for accuracy indicate differences between roof types (source: author’s own elaboration).

Metric	Roof Type	Mean	Standard Deviation	F-Statistics	p-Value
Precision	Gable	0.79	0.03	2.89	0.07
	Others	0.76	0.04		
Recall	Gable	0.78	0.03	2.65	0.08
	Others	0.75	0.04		
F1-score	Gable	0.78	0.03	2.77	0.07
	Others	0.74	0.04		
Total result of total	Precision	Recall	F1-score		
Total result of total	0.79	0.78	0.78		

The results of the ANOVA analysis for precision indicate that the differences between the roof types are on the borderline of statistical significance, as evidenced by the values $F = 2.89$ and $p = 0.07$ (Table A1). The precision of the trained YOLO model for gable roofs is 0.79, which is slightly higher compared with the other roof types (0.76). This suggests that the model is more effective in detecting objects of this class.

A similar result was obtained for the recall metric, with an ANOVA statistic of $F = 2.65$ and a p -value of 0.08. The model performs better at detecting gable roofs (recall = 0.78) than other roofs (0.75). These values confirm the slightly better ability of the model to detect objects with more regular geometry.

The F1-score measure also showed differences between the groups ($F = 2.77$, $p = 0.07$), with a score of 0.78 for gable roofs and 0.74 for others. This indicates an overall higher effectiveness of the model in detecting gabled roofs, a result of a balance between precision and sensitivity for this class of site.

The ANOVA performed allowed a detailed evaluation of the YOLO model, revealing differences in the detection performance of the two roof types. These results suggest the need for further fine-tuning of the model and the need to supplement the training data with examples of the other roof types. Marginalizing these differences would produce more consistent results and improve the overall performance of the model.

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